Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agriculture

Covers syllabus (click to expand)

Saponification & Detergent Chemistry (Micelle Dynamics, Hard Water Scum vs Synthetic Surfactants) | Pharmaceutical Chemistry (Analgesics, Antipyretics, Antiseptics vs Disinfectants, Antibiotics, Antacids) | Food Chemistry (Preservatives, Antioxidants, Sweeteners) | Industrial Glass Chemistry (Crookes UV Glass, Pyrex Borosilicate, Optical Flint Glass, Annealing) | Portland Cement Manufacture & Setting Chemistry | Agricultural Fertilizers (Urea 46% N, Neem-Coated Urea, DAP, Superphosphates, MOP) | Soil Chemistry (Liming Acid Soils, Gypsum for Alkaline Sodic Soils) | Agrochemicals (Herbicides, Insecticides, DDT Biomagnification, Zinc Phosphide Rat Poison) | Photography Chemistry (Silver Bromide, Hypo Fixer) & Artificial Rain (Silver Iodide)

Sources baked in: NCERT Class 10 & 12 (Chemistry in Everyday Life), ICAR Agriculture Chemistry Fundamentals, Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Urea nitrogen percentage (46%), Crookes glass UV absorption, Silver Bromide in photography vs Silver Iodide in artificial rain, Sodium chlorate as herbicide, Zinc phosphide as rodenticide, DDT as non-biodegradable pollutant, and Dettol components. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Nano Urea & Nano DAP Expansion**	IFFCO commercialization of **Nano Urea** (4\%\text{ N} w/v liquid nanoparticles, 20–50\text{ nm}) and **Nano DAP**: Provides >80\% nutrient uptake efficiency via foliar stomatal absorption, eliminating massive nitrogen leaching into groundwater and cutting conventional bagged urea consumption.
2024	**UPPCS Prelims 2024**	Question on **Photography Salt vs Artificial Rain**: Highlighted the recurring pair: **Silver Bromide (AgBr)** is the light-sensitive photographic emulsion salt, whereas **Silver Iodide (AgI)** is the cloud-seeding agent for artificial rain.
2023	**UPPCS / UKPCS Trends**	Persistent testing of **DDT (Dichlorodiphenyltrichloroethane)**: Identified strictly as a **non-biodegradable organochlorine pollutant** that undergoes severe trophic biomagnification in food chains (thinning avian eggshells).
2022	**UPPCS Prelims 2022**	Re-tested **Crookes Glass**: Keyed as the specialty glass containing Cerium oxide (CeO_2) that sharply cuts off and **absorbs ultraviolet (UV) radiation** in sunglasses. Pyrex/Borosilicate was an incorrect distractor.
2021	**UPPCS Prelims 2021**	Agricultural Herbicide: Keyed **Sodium Chlorate (NaClO_3)** and **2,4-D** as non-selective and selective weedicides/herbicides, respectively.

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master Classification of Industrial Glasses & Optical Formulations | Glass Variety Name | Fundamental Chemical Composition | Distinctive Physical / Optical Property | Primary Real-World & Applications | |---|---|---|---| | **Soda-Lime Glass (Soft Glass)** | $Na_2O \cdot CaO \cdot 6SiO_2$ ($\approx 75\% SiO_2$, 15\% Na_2O, 10\% CaO) | Low melting point, easily blown and molded | Domestic window panes, beverage bottles, glass tumblers, tableware | | **Potash Glass (Hard Glass)** | $K_2O \cdot CaO \cdot 6SiO_2$ | High softening temperature, chemical resistance | Laboratory combustion tubes, chemical reaction vessels | | **Pyrex / Borosilicate Glass** | $SiO_2\text{ (80\%)}$, $B_2O_3\text{ (13\%)}$, $Al_2O_3\text{ (3\%)}$, $Na_2O\text{ (4\%)}$ | **Low thermal expansion coefficient**, resists thermal shock | Chemistry beakers, flasks, pipettes, microwave oven bakeware | | **Flint Glass (Optical / Lead Glass)** | $K_2O \cdot PbO \cdot 6SiO_2$ (contains 20–30\% PbO) | High density, **high refractive index & high dispersion** | Optical camera lenses, microscope objectives, prisms, crystal chandeliers | | **Crookes Glass** | Soda glass containing **Cerium Oxide (CeO_2)** | **Selectively absorbs harmful ultraviolet (UV) rays** | High-grade UV-absorbing sunglasses, protective industrial welding goggles | | **Photochromic Glass** | Glass containing dispersed microcrystals of **AgCl or AgBr** | Darkens in bright sunlight (UV photolysis); clears indoors | Variable-tint prescription eyeglasses, solar sunglasses | | **Safety / Laminated Glass** | Two glass sheets sandwiching a layer of **polyvinyl butyral (PVB)** | Shatterproof; shards remain adhered to plastic interlayer | Automotive windshields, bullet-resistant security bank windows | --- ### Table 2: Master Commercial Fertilizer Matrix (NCERT / ICAR Standard) | Fertilizer Name | Primary Chemical Formula | Nutrient Content (% by weight) | Chemical Reaction in Soil | Special Agronomic Facts & Notes | |---|---|---|---|---| | **Urea** | **NH_2CONH_2** | **$46\%\text{ Nitrogen (N)}$** | Slightly acidic over long periods | Most widely used fertilizer in India; 100% produced as **Neem-Coated Urea**. | | **Ammonium Sulphate** | $(NH_4)_2SO_4$ | $21\%\text{ N}$, $24\%\text{ S}$ | Strongly Acidic | Ideal for waterlogged paddy (rice) crops; supplies essential sulphur. | | **Calcium Ammonium Nitrate (CAN)** | $CaCO_3 \cdot NH_4NO_3$ | $25\%\text{ N}$ | **Strictly Neutral** | Known as **"Kisan Khad"**; safe for all soils; non-acidifying. | | **Single Superphosphate (SSP)**| $Ca(H_2PO_4)_2 + 2CaSO_4$ | **$16\%\text{ P}_2\text{O}_5$, $11\%\text{ S}$, $19\%\text{ Ca}$** | Neutral / Slightly acidic | Ideal for oilseeds and pulses needing phosphorus and sulphur. | | **Triple Superphosphate (TSP)**| $3Ca(H_2PO_4)_2$ | **$46–48\%\text{ P}_2\text{O}_5$** | Slightly acidic | High-analysis phosphatic fertilizer without gypsum byproduct. | | **Diammonium Phosphate (DAP)** | $(NH_4)_2HPO_4$ | **$18\%\text{ N}$, $46\%\text{ P}_2\text{O}_5$** | Alkaline initially, then acidic | Second most widely consumed fertilizer in India after urea. | | **Muriate of Potash (MOP)** | KCl | **$60\%\text{ K}_2\text{O}$** | Neutral | **100\% imported into India**; unsuitable for chlorine-sensitive tobacco/potatoes. | | **Sulphate of Potash (SOP)** | K_2SO_4 | $50\%\text{ K}_2\text{O}$, $17.5\%\text{ S}$ | Neutral | Premium fertilizer for chloride-sensitive crops (tobacco, grapes). | --- ### Table 3: Agrochemical Classes & Target Pests | Agrochemical Type | Classic Active Chemical Compounds | Target Organisms / Ecological Action | Regulatory & Environmental Status | |---|---|---|---| | **Insecticides (Organochlorines)** | **DDT**, **BHC / Lindane**, Endosulfan, Aldrin | Controls mosquitoes, agricultural insect pests | **Non-biodegradable; biomagnifies in raptors**; banned globally under Stockholm POPs. | | **Insecticides (Organophosphates)** | Malathion, Parathion, Chlorpyrifos | Acetylcholinesterase inhibitors in insect nerves | Biodegradable but high acute mammalian toxicity. | | **Selective Herbicides** | **2,4-D (2,4-dichlorophenoxyacetic acid)** | Destroys broad-leaved dicot weeds in monocot cereals | Synthetic auxin mimicking growth hormones. | | **Non-Selective Herbicides** | **Sodium Chlorate ($NaClO_3$)**,

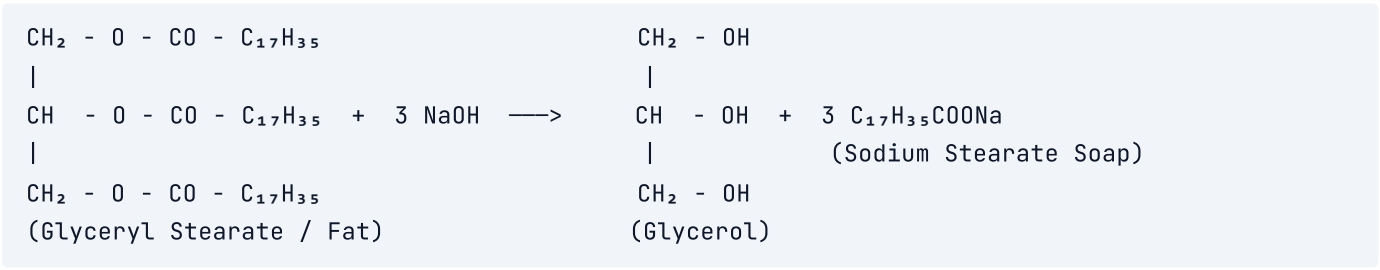
Glyphosate | Kills all green plant vegetation indiscriminately | Used for clearing railway tracks, industrial yards, weed knockdown. | | **Rodenticides (Rat Poisons)** | **Zinc Phosphide (Zn_3P_2)**, Warfarin | Stomach poison releasing lethal **Phosphine (PH_3)** | Standard acute poison for burrowing field rats. | | **Fungicides** | **Bordeaux Mixture ($CuSO_4 + Ca(OH)_2$)**, Sulphur | Copper ions destroy fungal spores | Standard spray for grape downy mildew and potato blight. | | **Cloud-Seeding Nucleants** | **Silver Iodide (AgI)**, Dry Ice (solid CO_2) | Hexagonal crystal template for ice crystal growth | Deployed by aircraft for artificial precipitation. | | **Photographic Salts** | **Silver Bromide (AgBr)**, Hypo ($Na_2S_2O_3$) | Light-sensitive emulsion; hypo fixes image | Foundational to classical monochrome photography. |

Detailed Theory Notes

7.1 Cleansing Chemistry: Saponification, Micelles & Synthetic Surfactants

1. Saponification Reaction

Soaps are manufactured by heating natural fats and oils (esters of glycerol with long-chain fatty acids) with aqueous alkali:



- The byproduct **Glycerol** is separated by fractional vacuum distillation and used in pharmaceuticals, cosmetics, and nitroglycerin explosives.

2. Micelle Mechanics

- The hydrophobic hydrocarbon tail is lipophilic (attracted to fats, repelled by water).
- The hydrophilic carboxylate head is ionic and solvated by water molecules.
- When concentration exceeds the **Critical Micelle Concentration (CMC)**, molecules self-assemble into a spherical cluster of 100–200 molecules.
- The non-polar dirt/grease is entrapped within the hydrophobic interior core. The charged carboxylate heads on the outer shell repel other micelles, preventing coalescence and keeping the oil droplets suspended as an emulsion that washes away with water.

7.2 Pharmaceutical Chemistry & Antiseptic Physics

1. Analgesics & Antipyretics

- **Aspirin (Acetylsalicylic Acid)**:
- Synthesized by acetylating salicylic acid with acetic anhydride in the presence of sulphuric acid: $C_6H_4(OH)COOH + (CH_3CO)_2O \to C_6H_4(OCOCH_3)COOH + CH_3COOH$

- Inhibits the enzyme cyclooxygenase (COX), blocking the biosynthesis of **prostaglandins** (pain and fever mediators) and **thromboxanes** (blood clotting agents).
- Used in low doses (75–150\text{ mg/day}) as an antiplatelet prophylactic to prevent coronary thrombosis and secondary strokes.

2. Antiseptic vs. Disinfectant Chemistry

- **Dettol**: The active germicide is **Chloroxylenol (4-chloro-3,5-dimethylphenol)**, blended with \alpha-terpineol in pine oil and soap solution. It disrupts bacterial cell membranes without irritating human skin.
- **Phenol Dichotomy**: Phenol (C_6H_5OH) was the historic antiseptic introduced by Joseph Lister (1865). At dilute concentrations (0.2\%), it denatures bacterial enzymes safely on skin. At higher concentrations (1.0\%), it damages human epithelial tissue and is restricted to disinfecting non-living floors and drains.

7.3 Industrial Glass Chemistry & Portland Cement Mechanics

1. Specialty Glasses

- **Crookes Glass**: Doping molten silicate glass with CeO_2 introduces localized electron transitions that absorb electromagnetic radiation with wavelengths below 400\text{ nm} (ultraviolet spectrum) while transmitting visible light (400–700\text{ nm}).
- **Pyrex (Borosilicate)**: Incorporating B_2O_3 reduces the coefficient of thermal expansion from 9 \times 10^{-6}\text{ K}^{-1} (soda glass) to 3.3 \times 10^{-6}\text{ K}^{-1}. When heated unevenly, thermal expansion differentials are minimal, preventing mechanical stress cracks.

2. Portland Cement Hydration & Setting

- When cement is mixed with water, hydration reactions form interlocking needle-like crystals of calcium silicate hydrate (C-S-H gel) and calcium hydroxide (Ca(OH)_2): 2(3CaO \cdot SiO_2) + 6H_2O \to 3CaO \cdot 2SiO_2 \cdot 3H_2O\text{ (C-S-H gel)} + 3Ca(OH)_2
- Gypsum (CaSO_4 \cdot 2H_2O) retards hydration of tricalcium aluminate (C_3A), ensuring a controlled setting profile.

7.4 Agricultural Fertilizers & Soil Remediation Chemistry

1. Urea and Nitrogen Dynamics

- Urea (NH_2CONH_2, M_w = 60\text{ g/mol}) contains two nitrogen atoms: \% N = \frac{2 \times 14}{60} \times 100 = 46.67\% \approx 46\%
- **Enzymatic Soil Hydrolysis**: NH_2CONH_2 + 2H_2O \xrightarrow{\text{urease}} (NH_4)_2CO_3 \rightleftharpoons 2NH_4^+ + CO_3^{2-}
- **Nitrification by Soil Microbes**: 2NH_4^+ + 3O_2 \xrightarrow{\text{Nitrosomonas}} 2NO_2^- + 4H^+ + 2H_2O 2NO_2^- + O_2 \xrightarrow{\text{Nitrobacter}} 2NO_3^-\text{ (absorbed by roots)}

2. Remediation of Problem Soils

- **Acidic Soils**: High rainfall leaches basic cations (Ca^{2+}, Mg^{2+}, K^+), leaving soil dominated by H^+ and Al^{3+}. Corrected by liming with $CaCO_3$ or $Ca(OH)_2$: $Al^{3+}\text{-Clay} + 3Ca(OH)_2 \to Ca^{2+}\text{-Clay} + Al(OH)_3\downarrow$
- **Alkaline / Sodic (Usar) Soils**: Arid evaporation concentrates exchangeable sodium (Na^+) on clay particles, causing soil swelling and poor drainage ($\text{pH} > 8.5$). Corrected with Gypsum ($CaSO_4 \cdot 2H_2O$): $2Na^+\text{-Clay} + CaSO_4 \to Ca^{2+}\text{-Clay} + Na_2SO_4\text{ (soluble, flushed out by irrigation)}$

Bilingual Terminology -- Chemistry in Daily Life and Agriculture

English Term	Hindi Term	Key Anchor
Fertiliser (उर्वरक)	रासायनिक उर्वरक	Chemical compound supplying nutrients to plants (N, P, K)
Urea (यूरिया)	यूरिया	Most used nitrogenous fertiliser; CO(NH2)2; 46% N
DAP (डीएपी)	डाईअमोनियम फॉस्फेट	Diammonium Phosphate; N + P fertiliser
Pesticide (कीटनाशक)	कीटनाशी / कीटनाशक	Chemical that kills pests; insecticide, fungicide, herbicide
DDT (डीडीटी)	डीडीटी	Dichlorodiphenyltrichloroethane; organochlorine insecticide; banned (persistent)
Insecticide (कीटनाशी)	कीटनाशी	Kills insects; e.g., malathion, BHC, pyrethroids
Herbicide (शाकनाशी)	शाकनाशी	Kills weeds; e.g., 2,4-D; Agent Orange (2,4,5-T + 2,4-D)
Fungicide (कवकनाशी)	फफूंदनाशक	Kills fungi; e.g., Bordeaux mixture (CuSO4 + lime)
Preservative (परिरक्षक)	परिरक्षक	Food additive preventing spoilage; salt, sugar, vinegar, sodium benzoate
Food Additive (खाद्य योज्य)	खाद्य योजक	Substance added to food for colour, flavour, preservation, nutrition
Dye (रंजक)	रंजक	Coloured substance; synthetic (aniline dyes) or natural (indigo, turmeric)
Soap (साबुन)	साबुन	Sodium/potassium salt of fatty acids; biodegradable cleansing agent
Detergent (अपमार्जक)	अपमार्जक	Synthetic cleaner; sulphonate/sulphate; works in hard water
Antiseptic (रोगाणुनाशक)	रोगाणुनाशक	Kills microorganisms on living tissue; dettol, savlon, iodine
Disinfectant (विसंक्रामक)	विसंक्रामक	Kills microorganisms on inanimate surfaces; bleach, phenol

English Term	Hindi Term	Key Anchor
Drug / Medicine (औषधि)	औषधि / दवा	Chemical compound used to treat, cure, or prevent disease

One-Liner Revision -- Chemistry in Daily Life and Agriculture

#	Fast Fact
1	Urea (CO(NH2)2) = most widely used nitrogen fertiliser; 46% N; water-soluble
2	DAP = Diammonium Phosphate; provides both N and P to crops
3	NPK fertilisers = provide Nitrogen, Phosphorus, Potassium (potash)
4	Bordeaux mixture = CuSO4 + Ca(OH)2; first effective fungicide; grape vines, potato blight
5	DDT = banned insecticide (persistent organic pollutant); biomagnification in food chain
6	Biomagnification = increasing concentration of persistent toxin up the food chain
7	BHC/BHC (Benzene Hexachloride) = lindane; insecticide; restricted use
8	Common antiseptics: Dettol (chloroxylenol), Savlon (chlorhexidine), Iodine (tincture)
9	Common disinfectants: bleaching powder, phenol (carbolic acid), formaldehyde
10	Paracetamol = analgesic (pain reliever) + antipyretic (fever reducer); most common OTC drug
11	Aspirin (acetylsalicylic acid) = analgesic, antipyretic, anti-inflammatory; anti-platelet
12	Antacids neutralise excess HCl: Mg(OH)2 (milk of magnesia), Al(OH)3, NaHCO3
13	Tranquilizers = CNS depressants; reduce anxiety; e.g., diazepam (Valium), barbiturates
14	Antibiotics kill bacteria; NOT effective against viruses
15	Antihistamines = counter histamine (allergic reaction); e.g., cetirizine, diphenhydramine
16	Soap = biodegradable; forms scum in hard water (Ca/Mg stearate precipitate)
17	Detergent = synthetic (non-soap); does NOT form scum in hard water
18	Food preservatives: salt (NaCl), sugar, vinegar (acetic acid), sodium benzoate, SO2
19	Food colours: Tartrazine (E102, yellow), Carmoisine (red) are synthetic; saffron, turmeric are natural
20	Baking powder = baking soda (NaHCO3) + cream of tartar (tartaric acid); releases CO2
21	Vinegar = 4-8% acetic acid (CH3COOH); food preservative and condiment
22	Chlorination of water = Cl2 or bleaching powder kills pathogens; residual Cl = 0.2-0.5 ppm
23	Fluoridation of water = NaF added; 0.7-1.2 ppm F- prevents tooth decay
24	Iodised salt = NaCl + KIO3 (potassium iodate); prevents iodine deficiency/goitre

#	Fast Fact
25	Organic farming = no synthetic pesticides/fertilisers; uses compost, biopesticides
26	Green Chemistry = design of chemical products that minimise environmental hazards
27	Food packaging: BPA (Bisphenol A) in polycarbonate plastics = endocrine disruptor
28	MSG (Monosodium glutamate) = flavour enhancer; "Umami" taste
29	Aspartame = artificial sweetener; ~200x sweeter than sugar; 0 calories
30	Saccharin = oldest artificial sweetener (1879); 300-500x sweeter; no calories